PDA3 Batch Analysis Report Symbol: ^GSPC

Source file: ^GSPC260101.csv

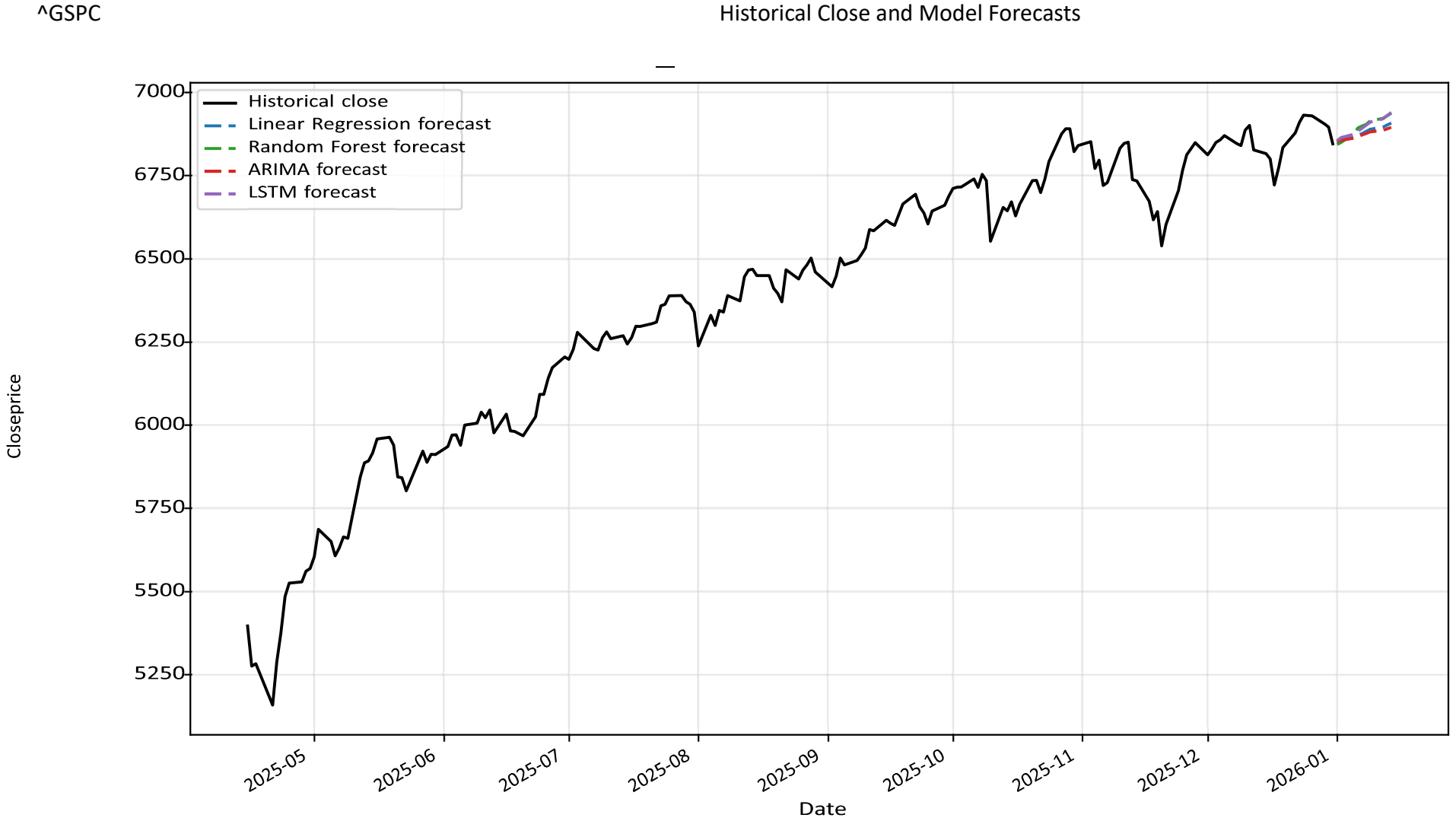
Generated: 2026-09-30 09:04 UTC

Observations used: 752

```
Latest indicators
  Last Close: 6845.5
  Change 30D: 2.59
  Sma 20: 6856.23
  Sma 50: 6802.55
  Volatility: 15.07
```

Estimates are educational model outputs, not investment advice. Historical accuracy does not guarantee future performance.

^GSPC


Historical Close and Model Forecasts
Historical close
Linear Regression forecast
Random Forest forecast
ARIMA forecast
LSTM forecast
7000
6750
6500
6250
6000
5750
5500
5250
Closeprice
2025-05
2025-06
2025-07
2025-08
2025-09
2025-10
2025-11
2025-12
2026-01
Date

Model validation metrics (walk-forward holdout)

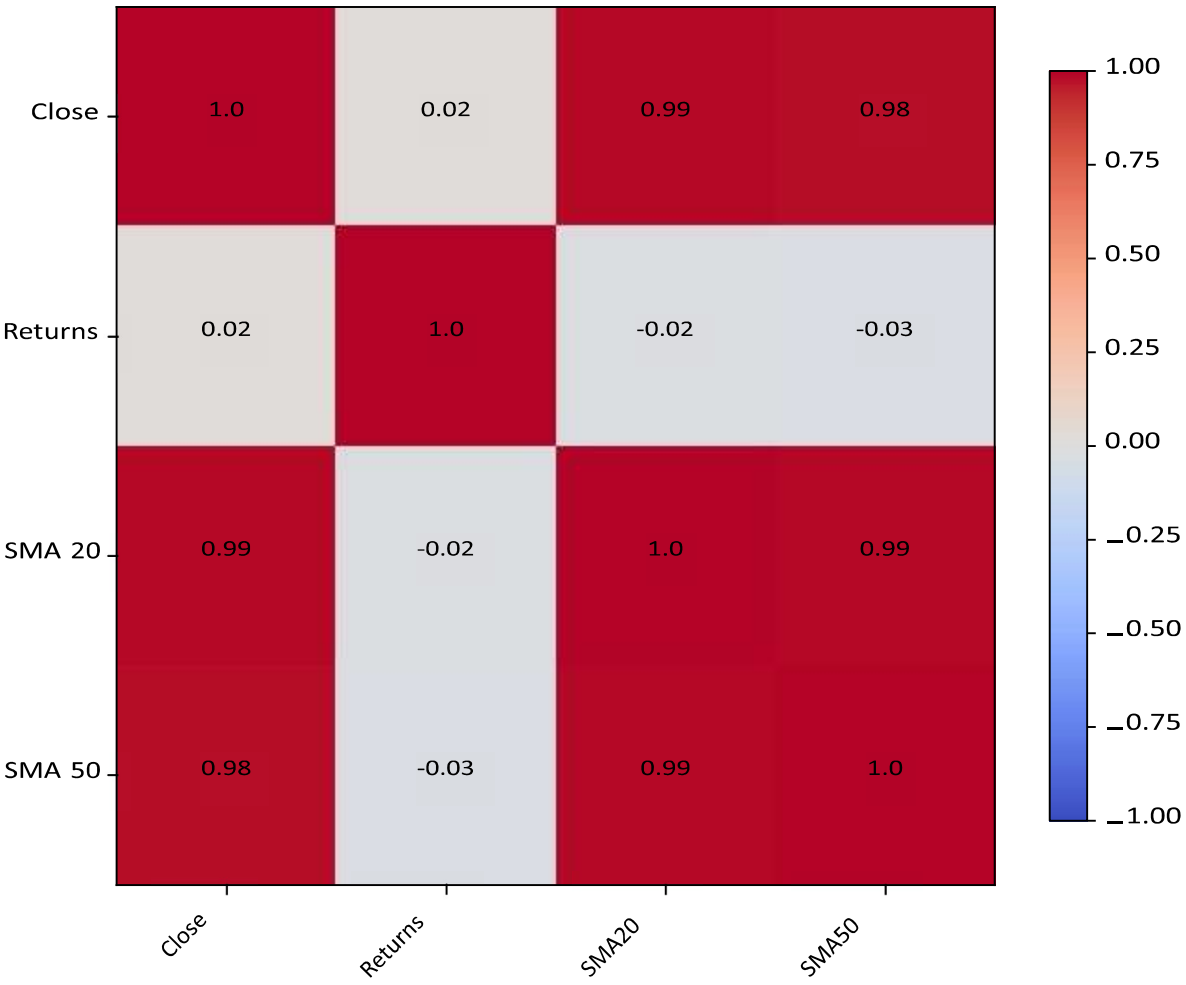
Model	MAE	RMSE	MAPE (%)	R^2	Notes
Linear Regression	193.00	234.14	2.68	0.6190	
Random Forest	231.129	278.8371	3.19	0.4575	
ARIMA	**156.09**	**222.49**	**2.23**	**0.6512**	
LSTM	468.2161	474.9818	7.65	0.514	

R^2 below 0 means a model performed worse than simply predicting the average price over the validation window common when a stock reverses or plateaus after a strong trend.

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Forecast table (next trading days)

Date	Linear Regression	Random Forest	ARIMA	LSTM
2026-01-01	6851.77	6844.02	6851.78	6854.92
2026-01-02	6858.03	6849.7	6856.62	6864.34
2026-01-05	6864.3	6875.25	6862.4	6873.76
2026-01-06	6870.56	6892.92	6866.93	6883.18
2026-01-07	6876.83	6899.01	6872.33	6892.6
2026-01-08	6883.1	6904.46	6876.65	6902.02
2026-01-09	6889.36	6915.04	6881.75	6911.44
2026-01-12	6895.63	6922.02	6885.92	6920.86
2026-01-13	6901.89	6930.14	6890.8	6930.28
2026-01-14	6908.16	6938.01	6894.87	6939.69



Correlation matrix

^GSPC Source data statistics (close price)

Statistic	Value
Count	752
Mean	5309.926
Median	5362.075
Std Dev	873.397
Min	3808.1
Max	6932.05
Range	3123.95
Q1 (25%)	4467.642
Q3 (75%)	5981.333
IQR	1513.69
Skewness	0.053
Kurtosis	-1.155
Coeff. of Variation (%)	16.448

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^GSPC Source data range and distribution

